**Why do People Listen to Sad Music?**

## Introduction

Why do people listen to sad music? This question is a bit of a paradox because the emotion of sadness has negative valence, and sadness is generally avoided; so why do people actively seek out sad music? Multiple hypotheses have been proposed to explain this phenomenon, dubbed the "sad music affinity paradox" in the literature. First, there might be psychological benefits that come from listening to sad music. If so, maybe there are specific situations – social, emotional, or otherwise – where the outcomes of listening to sad music are especially beneficial or help with self-regulation. Second, it is possible that there are specific auditory features associated with "sad" music that make it pleasing. In other words, is it truly the "sadness" of sad music which is sought or is it a different reaction evoked by the auditory features which are common to sad music?

We believe that it is worth understanding why people listen to sad music. If appreciation of sad music is linked to personality traits or mental health states, we could use sad music as a proxy to diagnose or identify them. In contrast, if appreciation of sad music is situational, elucidating this link could shed light on the role of sad music in emotional regulation. Finally, if it is the auditory features which are pleasing in sad music, discovering the essence of these features could allow us to generate music that is pleasing without the associated sadness.

## Research Background

The widespread enjoyment of sad music despite sadness being a negatively valenced emotion remains puzzling. It is widely recognized that people listen to music for emotional regulation (Sachs et al., 2015). Thus, it is perplexing that people would enjoy listening to sad music, as it is prone to induce negative moods. Hypotheses have been proposed to explain this phenomenon. One suggests that the aesthetics of the sad music itself are uniquely appealing (Schubert, 1996). Another suggests that sad music is particularly important for reflection on past experiences (Eerola & Peltola, 2016). However, few studies have provided a general predictive model for the specific situations in which sad music is enjoyed. There are 5 recognized parameters for the enjoyment of music. We aim to take a comprehensive approach investigating these parameters and their interactions. These parameters are: mood, personality, underlying mental conditions, familiarity to music, and musical features.

To begin with mood, as music is used for mood regulation, sad music may play an important role in maintaining emotional balance (Sachs et al., 2015). It is also possible that sad music can be used to contextualize a listener's emotions and past experiences (Schubert, 2016). Additionally, the aesthetic features of sad music have been shown to be helpful in obtaining desirable mood states such as peacefulness (Eerola & Peltola, 2016). However, bringing us to the

effect of underlying mental condition, mood effects may differ between people with conditions such as depression, ADHD or anxiety. One study shows that people who have been diagnosed as having major depressive disorder (MDD), have more preference for low energy sad music for calming and soothing effects to reduce their anxiety and mental pain (Yoon et al., 2020). Similarly, personality traits such as extroversion, openness, psychopathy and narcissism may affect the overall experience (Vuoskoski et al., 2012; Ladinig & Schellenberg, 2012). Thus we aim to investigate the effect of preexisting mental conditions, personality, and mood on the enjoyment of sad music.

These variables of psychological state also have interaction effects with familiarity of music and specific musical features. While previous studies have isolated one or more of these variables, nearly none have analysed their tandem effect. Beginning with familiarity, many studies have established the effect of familiarity through the difference between self-selected music, and unfamiliar music. Enjoyment of sad music has been previously noted to be affected far greater by familiarity than enjoyment of happy music (Weth et al., 2015). Additionally, liking of all music varieties is influenced by the quality of exposure to it with incidental exposure functioning very differently over time compared to focused exposure (Schellenberg et al., 2008). Finally, the last parameter – musical features – is important to discuss, though the literature on this subject is limited. Trained musicians are capable of recalling peak emotional responses to music and are able to identify musical features which cause these responses (Sloboda, 1991). This can be extrapolated to mean people have consistent responses to music features. Sad music generally has different musical features to music of other valences so this can be a reason why people may listen to it. For instance, music in minor key, which is often perceived as "sad", lends itself to a higher number of modal variations than major key.

To conclude, we aim to create a model that is effective in predicting enjoyment of sad music using these 5 parameters which have been shown to affect emotional response to music. From this model we can test the proposed hypotheses as to why people listen to sad music.

## Methodologies

### Participants

For this study, we will recruit a large and diverse sample of participants. We will need a large sample size in order for the study to be adequately powered (Wallisch, 2015). We will recruit participants both from the NYU subject pool (SONA) for course credit, and from the general community for monetary compensation. Participants will complete a pre-screening survey prior to starting the study to record demographic information, musical background, and self-reported mental health history.

### Materials and Measures

Each participant will listen to brief (5-second) clips of 400 songs, selected at random from a larger pool of 1,000 candidate songs that cover the entire range of music: Across time, genres

and popularity. The reason we believe that 5 second clips will evoke a sufficiently strong and reliable emotional response is that our lab demonstrated this to be the case previously (Philibotte et al., 2023). Each participant will rate their enjoyment of each clip on a 7-point Likert scale ranging from “Love it” to “Hate it”. In addition, each participant will also report which of 11 emotions known to be evoked by music (e.g. joy, groove, disgust, sadness, etc.) they actually experience. Each clip will be interspersed with a “palate cleanser” trial, either an assessment of current mood (every 80 clips) or an item from a personality instrument. This will help to minimize spillover effects between the clips. Importantly, it will allow us - en passant - to assess their personality, mood and mental health. Specifically, we will use items from the TIPI (Gosling et al., 2003), PHQ9 (Williams, 2014), Dirty dozen (Jonason & Webster, 2010).

**Procedure**

Our participants arrive at the testing location and are provided with a form to obtain written informed consent by research assistants. At this point, we will also confirm that they have completed our pre-screening survey. Once this is confirmed, they will be walked to the testing room, where they are seated at a computer and be given a set of headphones. Research assistants will demonstrate how the study works by going through a short training module and also set the sound to a comfortable volume. Research assistants then leave the participants to complete the study on their own, but are available outside the room, should there be any questions. After completing the study, participants are fully debriefed as to the purpose of the study.

**Analysis**

We will find clips that were experienced as sad by participants. We will further divide those clips into two “buckets”: Sad, but experienced positively (high rating) or sad, but experienced negatively (low rating). The question is whether we can find predictors of this bifurcated response pattern. Generally speaking, we will use a logistic regression model for binary classification to do so. In terms of predictors, we will first use Librosa (McFee et al., 2015) to characterize the audio features of the clips we used. If these are most predictive, the audio-feature model laid out in the introduction has the strongest empirical support.
We will use a similar approach with all of our other variables: Mood state, personality (particularly neuroticism) and mental health (particularly depression). This will allow us to assess which of these 5 parameters drive the enjoyment of sad music.

**Timeline**

Our research project is aimed to start in the first week of April and run through the first week of December 2026. Some of the data recordings are planned during this coming Summer semester of 2026. We anticipate about 10 hours per week for each member to be dedicated towards the work, with a total of 30 hours per week. This effort breaks down as follows: Each

week, we will record data from 10 participants, taking 20 hours - with no-shows, about 25 hours. The rest of the time, we will spend on data analysis and modeling. In November, the project will transition into full time writing and presentation mode, as we plan to present this work at a scientific conference and also prepare it for publication.

**Motivations**

This project is relevant to our academic interests and goals. Lorrie is a Computer Science major and Data Science minor, bringing together her technical expertise and her interest in music for our study. She has been interested in how data science tools can be applied to understand human behavior and culture, making this project a natural extension of her academic pursuits. Craig is a Data Science major and he is minoring in both Music and Math. Through this, he provides strong analytical skills and music domain knowledge. He is interested in pursuing an academic career so this project would be helpful in achieving that goal. Jenny majors in Data Science and minors in Child and Adolescent Mental Health Studies. She hopes to become a clinical psychologist who does empirical research, and this project will help broaden her research skillset by understanding psychology through a quantitative lens.

**Relationship with PI**

We took the class “Principles of Data Science” with Professor Wallisch in the Fall 2025 semester. We all very much enjoyed this class because of Professor Wallisch and his captivating teaching style. It inspired us to join his laboratory as research assistants. Since then, we have worked closely with Professor Wallisch to develop the idea for this project, and he has guided us every step of the way. He has helped us to work out the details of this study and provided his input on how to design this project. We have had biweekly meetings with Professor Wallisch to discuss our research project, where we receive guidance, feedback, and support. We will continue to do so as we move forward with the project.

**Costs**

As we aim for high statistical power (Wallisch, 2015), we will need to attain a large sample size. Whereas we will recruit some of our participants from the NYU SONA pool for class credit, it is also important to us to recruit a diverse pool of participants that are not NYU students. These participants will need to be paid at the rate of the minimum wage of $15 per hour in New York. Therefore, we request a budget of $1250 to recruit 40 participants, as each participant will have to be paid for 2 hours, allowing some money to pay for overtime, in cases where some participants will need more time, as our piloting suggests they will.
The payment of these participant fees is our only anticipated expense.

**Duties**

Every week, each member will assist in data recording for 6 hours, weekly lab meetings for 2 hours, and time allocated for data analysis of 2 hours each, summing up to a total of 10 hours. Data recording will be done in shifts of 2 hours by running participants that are scheduled at the start of each week. As every member has a strong background in Data Science and Computer Science, we will equally collaborate on data analysis to find out which of the hypotheses has the strongest empirical support. Thus, all team members will be equally contributing to the data analysis, but modulated by their interests and domain knowledge, for instance, Craig will lead the analysis of auditory features, Jenny will spearhead the analysis of psychological features and mental health status and Lorna will contribute by analyzing the impact of mood states and personality factors.

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